I. Setting up the Python environment and libraries

AIM:

To Create a new notebook for Python.To Create new cells for code and Markdown Demonstrate the application of Jupyter Widgets

PROCEDURE:

1. Open Jupyter Notebook or Google Colab and create a new Python notebook.
2. Import the required libraries, ipywidgets for creating interactive widgets and display from IPython.display for displaying them.
3. Create an integer slider using widgets.IntSlider() with a minimum value of 0, maximum value of 100, step size of 1, and an initial value of 50.
4. Create an output widget using widgets.Output() to display the slider value.
5. Define a callback function named on_value_change() to handle changes in the slider value.
6. Use output.clear_output() to clear the previous output whenever the slider value changes.
7. Display the updated slider value using a print() statement inside the output widget.
8. Register the callback function with the slider using slider.observe(on_value_change, names='value') so that the function executes whenever the slider value changes.
9. Finally, use display(slider, output) to display the interactive slider and its output in the notebook.
10. Execute the code cell and move the slider to different positions to observe the updated values.

PROGRAM:

```
import ipywidgets as widgets

from IPython.display import display

slider = widgets.IntSlider(value=50, min=0, max=100, step=1,
```

```
description='Value:')
output = widgets.Output()
def on_value_change(change):
    output.clear_output()
    with output:
        print(f'Slider value is {change["new"]}')
slider.observe(on_value_change, names='value')
display(slider, output)
```

OUTPUT:





RESULT:

Thus, the given program was executed successfully.